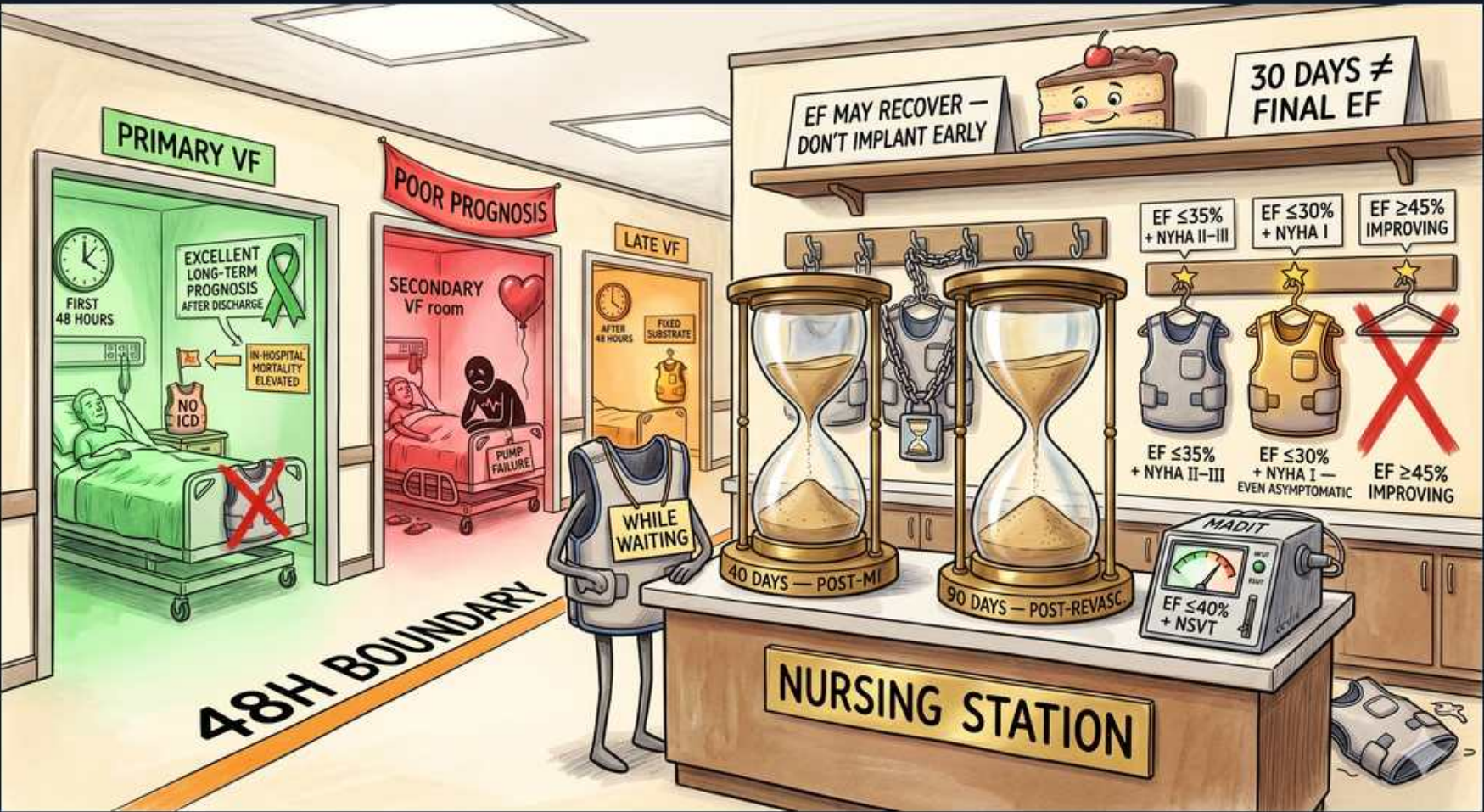


Post-MI Sudden Death — ICD Timing & EF Reassessment



1. PRIMARY VF room — first 48h

WHAT YOU SEE

Green ward labeled PRIMARY VF, clock 'FIRST 48 HOURS'.

WHAT IT ENCODES

Primary ventricular fibrillation occurs within the first 48 hours of acute MI, driven by ischemia/reperfusion. It carries an EXCELLENT long-term prognosis once revascularized and does NOT by itself indicate an ICD.

BOARD TRAP

VF in the first 48h of MI is NOT an ICD indication — it reflects acute ischemia, not a chronic arrhythmic substrate.

2. Green ribbon — excellent prognosis, no ICD

WHAT YOU SEE

Green ribbon, 'EXCELLENT LONG-TERM PROGNOSIS AFTER DISCHARGE', crossed-out ICD.

WHAT IT ENCODES

After early primary VF that is revascularized, long-term outcome is good and an implantable defibrillator is not indicated. The arrhythmia was a transient ischemic event, not a marker of permanent risk.

BOARD TRAP

Don't implant an ICD for primary VF — board answer is no ICD; the crossed-out ICD is the point.

3. POOR PROGNOSIS / SECONDARY VF room

WHAT YOU SEE

Red ward labeled POOR PROGNOSIS, 'SECONDARY VF', balloon.

WHAT IT ENCODES

Secondary VF (occurring with heart failure/cardiogenic shock during MI) signals a sicker heart and worse in-hospital mortality. It reflects pump failure substrate rather than transient ischemia alone.

BOARD TRAP

Secondary VF (hemodynamic instability/HF) has worse prognosis than primary VF — don't conflate the two.

4. LATE VF room — fast reassessment

WHAT YOU SEE

Ward labeled LATE VF with 'FAST REASSESSMENT'.

WHAT IT ENCODES

VF/VT occurring >48h after MI (not from reversible ischemia) reflects a true arrhythmic substrate and warrants evaluation for secondary-prevention ICD. Late sustained ventricular arrhythmia carries high recurrence risk.

BOARD TRAP

Late (>48h) sustained VT/VF is the dangerous one — it, not early primary VF, points toward ICD evaluation.

5. 48H BOUNDARY line

WHAT YOU SEE

Floor line labeled '48H BOUNDARY'.

WHAT IT ENCODES

The 48-hour mark divides ischemia-driven primary VF (good prognosis, no ICD) from later arrhythmias reflecting a fixed substrate. This temporal boundary is the core discriminator for post-MI sudden-death management.

BOARD TRAP

The whole decision hinges on timing relative to 48h — memorize the boundary, not just the EF cutoffs.

6. Hourglass — 40 days post-MI

WHAT YOU SEE

Hourglass labeled '40 DAYS POST-MI'.

WHAT IT ENCODES

Primary-prevention ICD for low EF after MI must wait at least 40 days post-MI (and ≥90 days post-revascularization). Early implantation (DINAMIT/IRIS trials) showed NO mortality benefit because EF often recovers.

BOARD TRAP

Do not implant a primary-prevention ICD before 40 days post-MI — early studies showed no survival benefit.

7. Hourglass — 90 days post-revascularization

WHAT YOU SEE

Second hourglass '90 DAYS POST-REVASC'.

WHAT IT ENCODES

After revascularization (PCI/CABG), wait ≥90 days before primary-prevention ICD, allowing myocardial stunning/hibernation to recover and EF to be reassessed at its true nadir.

BOARD TRAP

Two clocks: 40 days post-MI AND 90 days post-revascularization — the longer applicable wait governs the decision.

8. Cake / 'EF may recover — don't implant early'

WHAT YOU SEE

Cake icon, 'EF MAY RECOVER — DON'T IMPLANT EARLY'.

WHAT IT ENCODES

Post-MI EF is frequently depressed transiently and recovers with revascularization and guideline-directed medical therapy. Reassess EF after the waiting period before deciding on an ICD.

BOARD TRAP

A low EF measured during the acute MI does NOT qualify for an ICD — it must persist on REASSESSMENT after 40/90 days on optimal therapy.

9. '30 days ≠ final EF' sign

WHAT YOU SEE

Sign reading '30 DAYS ≠ FINAL EF'.

WHAT IT ENCODES

EF at 30 days is not the definitive value; reverse remodeling on GDMT continues. The reassessment EF after the mandated waiting window is what determines ICD eligibility.

BOARD TRAP

Don't lock in the early EF — the final, post-optimization EF is the one that counts for ICD.

10. EF ≤35% + NYHA II-III vest

WHAT YOU SEE

Gray vest 'EF ≤35% + NYHA II-III'.

WHAT IT ENCODES

Primary-prevention ICD is indicated for EF ≤35% with NYHA class II-III on ≥3 months optimal medical therapy (and past the post-MI/revasc waiting period). This is the broadest qualifying group.

BOARD TRAP

EF ≤35% alone is not enough — symptomatic class (NYHA II-III) and the waiting period must also be met.

11. EF ≤30% + NYHA I vest

WHAT YOU SEE

Yellow vest 'EF ≤30% + NYHA I'.

WHAT IT ENCODES

Even asymptomatic (NYHA I) patients qualify for a primary-prevention ICD if EF ≤30% (MADIT-II criterion) after the waiting period. The lower EF threshold captures very-high-risk asymptomatic patients.

BOARD TRAP

NYHA I patients DO qualify — but only at the stricter EF ≤30% cutoff, not ≤35%.

12. EF ≥45% improving — crossed out

WHAT YOU SEE

Red-X vest 'EF ≥45% IMPROVING'.

WHAT IT ENCODES

If EF recovers to ≥45% (improving) on therapy, no ICD is indicated — the patient no longer meets low-EF criteria. This is exactly why early implantation is avoided.

BOARD TRAP

Recovered EF (≥45%) means NO ICD — implanting based on the early low value would be wrong.
